

Review

Reviewed Work(s): Proceedings of the International Conference on Literary and Linguistic Computing, Israel by Zvi Malachi

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form of a story or essay. Although their introduction shows that they have a thorough grasp of Lear's writings and their most significant features, they seem to have intervened in favor of the general reader rather than the scholar. Thus they include an alphabetical index of first lines and titles but exclude a sorely needed ranking frequency table. They have worked hard at 'grammatical disambiguation' but have failed to distinguish Lear's coinages: *scroobious* follows *screwed* and precedes *scrunched* without having any special designation to set it apart as a Learian invention. One would like to see some symbol beside the nonsense words, or at least a list of them in the back matter.

Perhaps the exigencies of publishing rather than deficiencies in the compilers account for these omissions. They anticipate many of the above criticisms in the introduction and state that, happily, interested scholars may obtain the unedited computer text – 250 pages longer – and special tables through the Center for Computer Research in the Humanities at the University of Colorado at Boulder. What the published version does contain is presented in an accessible format and an attractive camera-ready typeface. The context lines are wisely chosen; the abbreviations of poem titles and dates, and the page references to the Holbrook Jackson edition of *The Complete Nonsense* that stand to the left of each entry make location of the source quick and easy.

For anyone contemplating serious work on Edward Lear, this concordance, even with its flaws, is a must.

Proceedings of the International Conference on Literary and Linguistic Computing, Israel, Ed. Zvi Malachi, Tel Aviv, 1979.

By L.D. Burnard

For one week at the end of April 1979 the Katz Institute of the University of Tel Aviv was host to a special conference organized under the aegis of the

ALLC. There were 38 participants, half of them from Israel, with smaller delegations from the Netherlands, the U.K. and the U.S.A. together with a handful of onlookers from the rest of Europe and one solitary Australian. 48 papers were presented, two-thirds of these being reproduced in the present volume. The academic receptions, cocktail parties, sight-seeing tours and other customary junketings are also noted therein, which is perhaps not inappropriate. For the function of this volume (I hope its editors will forgive me for dwelling on the point) is more to recall an occasion than to contribute overmuch to the state of knowledge. More souvenir than symposium, it resembles nothing so much as the contents of a particularly zealous delegate's briefcase on his return home. Here is Professor X's abstract, promising the philosopher's stone but not delivering it for lack of space – rather like Fermat's last theorem; here is Professor Y's sheaf of smudgy printout, demonstrating his newfound skill as a programmer; here is Professor Z's sales brochure for some new piece of exotic computer software. Above all here is evidence that the on-going preoccupations of the University of Laputa's Special Research Section on Computer Aided What Not are still going on, after a decade or more of faithful attendance at international symposia.

In the circumstances it is not surprising that it should be the home team which fields the strongest side in this particular international encounter. Few would disagree that the production of the Historical Dictionary of the Hebrew Language is one of the most ambitious projects of modern lexicographical scholarship, computer-aided or otherwise. The sheer bulk of materials it is to describe – over two millennia of linguistic production – is awe-inspiring enough in itself, without the added complications inherent in the Hebrew language. As Professor Ben-Hayyim, the President of the Academy of the Hebrew Language, remarks in his opening address, the project has had to find its own way towards solving methodological problems "to which no solution had previously been sought". This is not of course to deny the value of earlier dictionaries (most notably Ben Yehuda's *Thesaurus totius Hebraicitatis*) but simply to demonstrate how rapidly a difference of quantity fades into a difference of quality. Regrettably, not all of the Israeli contributions describing work in this field

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are here reproduced; particularly conspicuous by their absence are any account of the Responsa Retrieval Project (which has however been described in a recent report in *Computers and the Humanities*) or of the Complete Yiddish Dictionary Project. Of those which are reprinted, however, three papers, those by Yeivin (on the methods used in pre-editing ancient texts for lexicographical analysis), Adler (on the implementation of an online Hebrew thesaurus) and Busharia (on methods of disambiguating unpointed Hebrew), are particularly worthy of attention by anyone interested in the problem of exactly how such tasks may be performed. It is also worth noting that the method evolved by Busharia and his co-workers for disambiguating Hebrew has much in common with the methods used by the Norwegians for lemmatizing Ibsen; whether the same method was used for lemmatizing the Belgian mediaeval Latin corpus described in this volume by Tombeur is less clear. Of course the method is hardly novel (it has been in use for ten years or more) but it has the great merit of being over 90% accurate, unlike that presented in another paper on lemmatisation by Goldberg and Choueka. They take a more 'information-theoretic' approach, presenting some highly sophisticated algorithms and measures of performance based on four rather dubious axioms concerning the relation between wordform ("word") and wordroot ("meaning"), with satisfyingly disappointing results.

There is a great variety amongst the other contributions to this volume, reflecting no doubt the diversity of membership enjoyed by the ALLC. Thus we find cheek-by-jowl one article enthusiastically describing how paper tape and Snobol may be used to generate Russian printout and another in which are expounded the pros and cons of a state-based representation of adverbial modifiers in an expanded semantic network. Such eclecticism, however admirable, runs the serious risk of becoming an uncritical acceptance of any p-baked ($0 < p < 1$) idea or project in which counting, logical formalisms or algorithmic methods are involved. Selection is individual, but two extreme examples of the type of article which I have in mind may be cited: one by Tuttle on the counting of suffixes as a test for neologisms in Rabelais and one by the perhaps better-known Melčuk on the "Meaning $\langle \Rightarrow \rangle$ Text" Model

of language. Tuttle is merely incomprehensible (a prize of 50p is hereby offered for the best explanation of the term "non-parallellicity of non-trivial total graph" which appears throughout his article) while Melčuk transcends comprehensibility in what I cannot resist dubbing a TBG (Tantalizingly Brief Glimpse) of the new SemRs (Semantic Representations) possible within "LMs (linguistic Models) of the Meaning $\langle \Rightarrow \rangle$ Text Type (MT models or MTMs)".

Cercone and Slaby both report on projects already described in the proceedings of the previous ALLC Conference (*Advances in Computer Aided Literary and Linguistic Research*, ed. Ager, Knowles and Smith; Aston, 1979); Slaby's article is more or less identical in the two publications, and Cercone's differs only in the stress given different aspects of the AI system described. The figures are better reproduced in the Israeli version, but the bibliography is missing, which is a pity, as it is a useful source of authoritative information about expanded semantic nets. Using another well-beloved formalism, Sallis attempts to represent formal argumentative prose by a context-free grammar. This attempt must have delighted Boot (whose paper here proposes such grammars as the new *lingua franca* of text analysis software) as much as it depresses this reviewer. A context-free argument is, I contend, no argument at all but an algorithm; irony has no place in such an analysis and is yet one of the most powerful tools of argumentative writing (see Milton's prose, *passim*). Marinov's paper seems to be SHRDLU revisited, but is well written and clearly argued enough to demonstrate the limitations of any system in which procedural knowledge cannot be acquired but must be represented.

Of more practical interest are two papers which attempt to bridge the gap between data analysis and query processor: King's evaluation of three well-known database systems in terms of their using (or not using) the semantic information held within the database structure, which opens up a promising line of research; and Krause's account of USL, a query-processing system with a powerful linguistic component. As is often the case, however, more effort seems to have gone into evaluating the system than designing it. Tapper proposes another type of approach to the general problem of rendering accessible the information held in those giant databases

of which the media have such horror: the use of citations rather than keyword as primary index to legal databases has much to attract the busy barrister but whether it is a technique of general applicability is more debatable. However, the popularity of such reference works as *Science Citation Abstracts* perhaps indicates that my doubts are ill-founded.

Several papers deal with that branch of analysis which has acquired the name of stylometry, after the assiduous promulgation of the term by Morton and his followers. This technique, as Bailey's magisterial recension of Kenny's work on Aristotle correctly points out, remains a science in search of a theory. We still have no adequate theoretical model of linguistic behavior, and the process by which variables are selected to make one up inevitably seems somewhat arbitrary. Greatly needed is some method by which the seemingly *ad hoc* collection of variables available for statistical analysis of text can be clustered, discriminated and filtered. In this connection, two papers, one by Weil and another by Shore and Radday, touch upon the use of factor analysis, the latter rather more thoroughly than the former. Both papers deal with attempts to find quantitative evidence to support the seemingly intuitive and qualitative groupings of the books of the Bible which have been inherited from generations of biblical scholars, but neither can be said to be entirely successful.

Schore & Radday's case is perhaps more clearly argued, as is their method, but neither paper tackles the fundamental applicability of techniques which have been evolved for parametric statistics to statistics such as word frequency. It may be claimed that the relative frequencies of some syntactic markers or of high-frequency function words are (within the limits of experimental error) 'interval scaled', but with no theoretical underpinning to explain why this should be, the sceptic will remain unconverted. In particular, he will assert that the difference between a word frequency of 0 and a word frequency of 1 is not the same as that between frequencies of say 10 and 11, and that therefore word frequency is not strictly speaking a parametric statistic at all. Hamesse's paper on an attempt to identify an author's 'preferred language' independent of *topos*, by inspection of the relative frequencies of non-function words, is even more suspect. 'Least squares' analyses, where such tests as Student's *t* are used to establish degrees

of significance, are worthless when applied to variables which are not normally distributed. Whatever else it may be, the distribution of lexical tokens in the works of St. Bonaventura (or of anyone else yet analyzed), particularly those of high semantic content, is not normal.

The volume also contains a number of contributions which might be dignified by the title of semi-semiotic analysis. Phelan on Chaucer's language is novel, if somewhat too enthusiastic and inaccurate for my taste. Weiss on the word "throne" in a short story by the Hebrew writer Agnon is more restrained in his language, but is still evidently reeling from culture shock. Bender on Conrad's vocabulary promises to substantiate one of the commonplaces of Conrad criticism, but does not deliver. This is a pity, because the proper study on vocabulary and in particular of vocabulary collocations should be an important adjunct to more purely 'literary' criticism. A recent work by Lakoff and Johnson proposes metaphor as a fundamental part of perception; we should expect to find its analysis at the heart of literary studies instead of, on the present showing, its lunatic fringes.

The volume has been assembled with commendable speed and a faithfulness to its authors' original typescripts, which this reviewer perversely refuses to commend. Leaving aside the somewhat pointless presence of a number of one-page abstracts, the volume crawls with elementary typing errors and linguistic solecisms which I believe it is the duty of an editor tactfully to correct. Of course such corrections would have delayed the publication of the work; they would also completely preclude a comparative study of the typing or linguistic skills of the distinguished contributors or a survey of the levels of secretarial technology at different universities. I count these small losses set against the number of times in this volume that a fairly murky argument is rendered utterly opaque by a simple error of transmission, an inverted page, or reference to a missing footnote.